

KUNAL KISHORE

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Education

University of California Los Angeles

Expected: Dec 2022

Master of Science in Electrical and Computer Engineering - Machine Learning Track

Indian Institute of Technology Madras | Chennai, India

May 2017

Bachelor of Technology - Major in Electrical Engineering, Minor in Operations Research

GPA : 8.72/10

Relevant Coursework

- *Course Plan at UCLA:* Neural Networks and Deep Learning, Large Scale Data Mining, Statistical Machine Learning, Convex Optimization, Digital Image Processing, Digital Speech Processing
- *Courses Completed:* Linear Programming, Introduction to Machine Learning, Matrix Analysis & Linear Algebra, Data Structures and Algorithms, Computer Architecture, Probability & Statistics, Adaptive Signal Processing

Technical Skills

- **Programming Languages:** Python, SQL, C++, Go, Bash Script, Javascript
- **Libraries/Frameworks:** Tensorflow, Pytorch, Pandas, NLTK, Scikit-Learn, Cvxpy, Numpy, SQLite
- **Developer Tools:** Git, Jupyter Notebook, MATLAB, VS Code, Source Insight, Perforce

Professional Experience

Samsung R&D Institute | Bangalore-India, Seoul-South Korea

Jul 2017 – Aug 2021

Lead Engineer

- Designed novel algorithm for Adaptive Resource allocation in downlink; improved spectral efficiency by **2.6%**.
- Proposed a design(pending patent) for enhancing state-of-the-art Carrier aggregation methods; increased data throughput by more than **50%**.
- Built a simulation tool on C++ framework for LTE MAC layer testing and development; sped up project release timeline by **a month** - received Spot Award for the same.

Samsung R&D Institute | Bangalore-India

May 2016 – Jul 2016

Software Intern

- Developed a Cognitive radio module for predicting active scheduling at the Mobile(UE) to save power.
- Achieved **4%** lesser battery power consumption at the cost of **just 0.5%** false detection rate.

Centre for Excellence in Wireless Technology | Chennai-India

May 2015 – Jul 2015

Software Intern

- Designed a 3D Ray Tracer model to check the feasibility of using Millimeter Waves for the 5G cellular systems.
- Reduced simulation run time by **90%** using Concurrent Programming in GoLang; verified it on HPCE cluster.

Projects

News Article Classification Pipeline | *Python*

Jan 2022 - Feb 2022

Course Project in Course: Large Scale Data Mining

UCLA

- Performed articles classification using TF-IDF feature extraction based on bag-of-words model; used PCA & NMF for dimensionality reduction before applying the SVM classification model.
- Replaced the Count Vectorizer with a Glove Embedding model and observed **3%** improvement in accuracy; used gridsearch for hyperparameter tuning to further increase performance by more than **2%**.

Edge Machine Learning using Sample Selection in SVM | *Python*

Sep 2021 - Dec 2021

Course Project in Course: Linear Programming

UCLA

- Implemented an online diverse sample selection algorithm; reduced input sample size by **100 fold** prior to invoking the SVM classifier; more than **90%** reduction in training duration.
- Developed an offline algorithm for the same task using KNN and K-Means based clustering for support vectors extraction; observed increase of **3%** in accuracy compared to online algorithm.

Fashion Item Classifier | *Python*

Jan 2022 - Current

Course Project in Course: Statistical Machine Learning

UCLA

- Implementing various multi-class classification models including Perceptron, Logistic Regression, SVM and Decision Tree with Adaboost on the Fashion-MNIST dataset.
- Comparing usage of Kernel functions in Linear classifiers against using a Neural Network directly on input data.